

Hybridisation: domains, geometry and model choice:

Source questions

Q1 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Hybrid set size

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Electron domains and introductory hybrid-set mapping

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

In the introductory valence-bond model, name the hybrid-orbital set used for four equivalent electron-domain directions

Representation

Start with direction count

3 directions

4

around a central atom, and state how many hybrid orbitals it contains.

Conditions

- Use the stated central atom and the introductory VSEPR/valence-bond model; count electron-density regions rather than drawn bond lines.

Common wrong route · open if you want a warning

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Domain count

Need a hint? · guided support

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Need the concept again?

Hybrid set size

Q2 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Count central electron domains

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Electron domains and introductory hybrid-set mapping

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

The Lewis structure $\text{H}_2\text{C}=\text{O}$ has a double bond from carbon to oxygen and one single bond from carbon to each hydrogen. Explain why the $\text{C}=\text{O}$ double bond counts as one electron domain around carbon.

Conditions

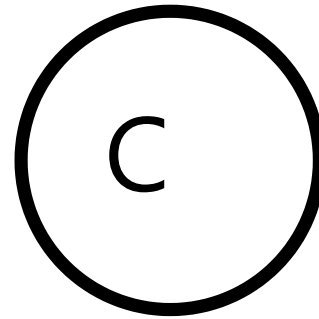
- Use the stated central atom and the introductory VSEPR/valence-bond model; count electron-density regions rather than drawn bond lines.

Common wrong route · open if you want a warning

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Representation

Choose the counting c



Choose centre

Need a hint? · guided support

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Need the concept again?

Count central electron domains

Q3 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Four domains to sp^3 and tetrahedral geometry

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Electron domains and introductory hybrid-set mapping

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For CH_4 , use the known rule for four bonding domains and no central lone pair to assign the carbon hybridisation and electron-domain geometry. Show the domain count.

Representation

Conditions

- Use the stated central atom and the introductory VSEPR/valence-bond model; count electron-density regions rather than drawn bond lines.

Common wrong route • open if you want a warning

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4 C–H bonding domains

C

Four bonding domains

Need a hint? · guided support

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Need the concept again?

Four domains to sp^3 and tetrahedral geometry

Q4 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Lewis structure to tetrahedral domain arrangement

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Lewis-to-3D domain geometry

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

NH₃ has three N–H single bonds and one lone pair on nitrogen. Translate that information into a labelled diagram showing all four electron domains; distinguish the lone pair from the three

Representation

bond directions. Do not treat the planar Lewis sketch as the molecular shape.

Conditions

- Use the stated central atom and the introductory VSEPR/valence-bond model; count electron-density regions rather than drawn bond lines.

Common wrong route · open if you want a warning

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NH₃: three N-H bonds; one N l

N

Inventory

Need a hint? · guided support

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Need the concept again?

Lewis structure to tetrahedral domain arrangement

Q5 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Choose a model by explanatory scope

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Model-scope selection

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A model is needed only to describe the overall three-dimensional arrangement of electron domains in NH_3 ; it need not describe the detailed orbital overlap. Choose between VSEPR and a hybrid-orbital overlap diagram, and identify what your chosen model predicts and what it leaves unresolved.

Conditions

- Choose the model only for the explanatory

Representation

Q5 asks for overall 3D electron-domain a

Requested output

Need a hint? · guided support

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Need the concept again?

Choose a model by explanatory scope

output requested in the stem.

Common wrong route · open if you want a warning

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Q6 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Sigma and pi overlap geometry

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Sigma/pi overlap geometry

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For a qualitative orbital sketch, overlap A is end-on along the internuclear axis and overlap B is side-on on opposite sides of that axis. Classify each as sigma or pi

Representation

internuclear axis

Internuclear axis

overlap and identify the defining feature you used.

Conditions

- Classify overlap from orbital orientation relative to the internuclear axis.

Need a hint? · guided support

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Need the concept again?

Sigma and pi overlap geometry

Common wrong route · open if you want a warning

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Q7 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Lewis structure to tetrahedral domain arrangement

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Lewis-to-3D domain geometry

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For CH₄, sketch a three-dimensional arrangement of the four C–H bond directions using a wedge/dash convention or another clearly defined convention.

Explain how your drawing differs from a flat cross-shaped Lewis sketch.

Conditions

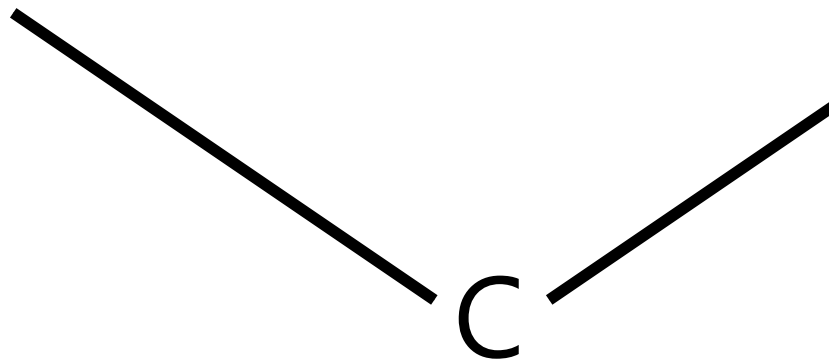
- Use the stated central atom and the introductory VSEPR/valence-bond model; count electron-density regions rather than drawn bond lines.

Common wrong route · open if you want a warning

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Representation

H



Connectivity

Need a hint? · guided support

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Need the concept again?

Lewis structure to tetrahedral domain arrangement

Q8 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Count central electron domains

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Electron domains and introductory hybrid-set mapping

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A nitrogen atom in NF_3 has three N–F single bonds and one lone pair. Each fluorine also has lone pairs. Count the electron domains around nitrogen and state which lone pairs belong in that central-atom count.

Conditions

- Use the stated central atom and the introductory VSEPR/valence-bond model;

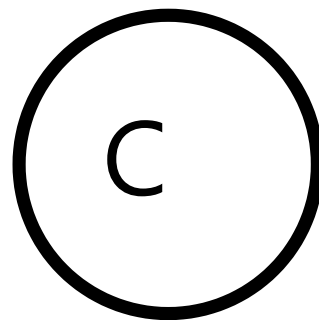
Representation

count electron-density regions rather than drawn bond lines.

Common wrong route · open if you want a warning

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Choose the counting c



Choose centre

Need a hint? · guided support

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Need the concept again?

Count central electron domains

Q9 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Count central electron domains

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Electron domains and introductory hybrid-set mapping

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A student counts the double bond in $\text{H}_2\text{C}=\text{O}$ as two separate directions and predicts four domains around carbon. Locate the counting error and give the corrected domain count.

Representation

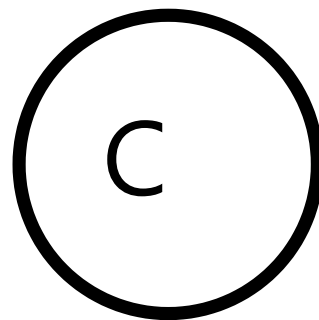
Conditions

- Use the stated central atom and the introductory VSEPR/valence-bond model; count electron-density regions rather than drawn bond lines.

Common wrong route · open if you want a warning

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Choose the counting c



Choose centre

Need a hint? · guided support

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Need the concept again?

Count central electron domains

Q10 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Four domains to sp^3 and tetrahedral geometry

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Electron domains and introductory hybrid-set mapping

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

The benchmark supplies this table: BF_3 has three bonding domains and no central lone pair; CH_4 has four bonding domains and no central lone pair. Match each central atom to the introductory sp^2 or sp^3 model,

Representation

Start with direction count

3 directions

4

and check that the number of hybrid-orbital directions agrees with the domain count.

Conditions

- Use the stated central atom and the introductory VSEPR/valence-bond model; count electron-density regions rather than drawn bond lines.

Common wrong route · open if you want a warning

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Domain count

Need a hint? · guided support

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Need the concept again?

Four domains to sp^3 and tetrahedral geometry